MC

SPECIFIC UNIVERSITY · DATA SCIENCE

Maya Chen: why Data Science

IB Diploma · Year 13 · for Northbridge University (fictional), BSc Data Science

To the admissions team, Northbridge University (fictional), BSc Data Science

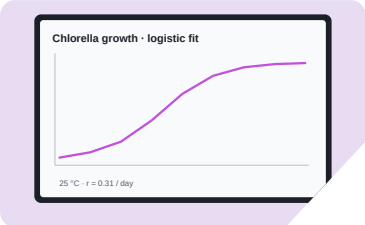
Why Data Science

Every line on my map that kept growing ended in a dataset. Data Science is the course that joins the questions I learned to ask in Biology with the tools I taught myself in Computing.

Maya Chen

EVIDENCE FOR THIS COURSE

Science · Biology



Biology IA: modelling algae growth 20 Apr 2026

Fitted a regression model to 12 weeks of growth data.

> *"Statistical treatment well beyond the usual IA."*
> **Mr David Okafor, Biology**

Science · Biology

Data Science

Teacher-endorsed `NIST-1TGTG6`

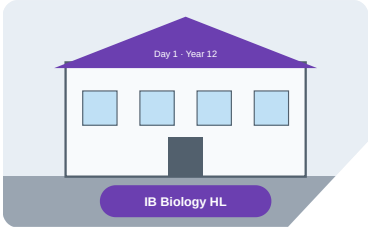


Biology IA proposal 20 Jan 2026

Proposal: algae growth against water temperature.

Science · Biology

School-verified `NIST-1SWXEA`



Started IB Biology HL 20 Aug 2025

Depot: IB Biology HL with Mr Okafor.

Science · Biology

School-verified `NIST-1VETTI`



Enzyme kinetics investigation

10 Mar 2025

Measured catalase activity across temperatures; first time graphing error bars.

> *“Exceptional control of variables and uncertainty.”*
> **Mr David Okafor, Biology**

Science · Biology

Teacher-endorsed `NIST-T437AV`



Genetics unit test

2 Nov 2024

Year 11 genetics test.

Science · Biology

School-verified `NIST-Q27K06`



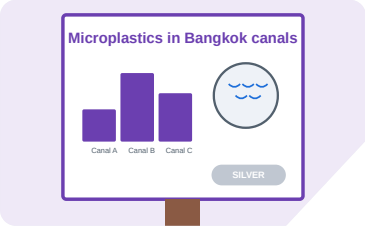
Reflection: why data keeps pulling me in

20 Jun 2024

End-of-year reflection on the science fair and the composting audit.

Science · Biology

Student-added



Science fair: microplastics

15 May 2024

Filtered water from three canals and counted microplastic fibres.

Science · Biology

Student-added



Cell biology lab report

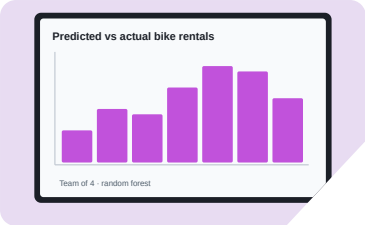
12 Oct 2023

Microscopy of onion and cheek cells; scored 7/8 on criterion B.

Science · Biology

School-verified `NIST-1QOU9Q`

Data Science



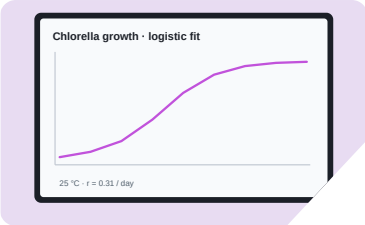
Data Science summer school

10 Jul 2026

Two-week online summer school; final project on bike-share demand.

Data Science

Student-added



Biology IA: modelling algae growth

20 Apr 2026

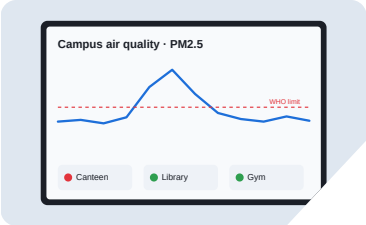
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> *“Statistical treatment well beyond the usual IA.”*
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Science · Biology

Data Science

Teacher-endorsed `NIST-1TGTG6`



Air-quality data report

28 Mar 2026

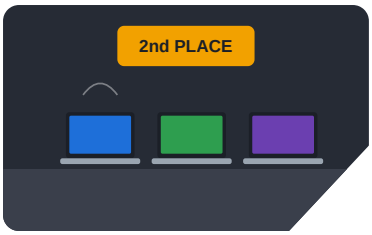
Cleaned 40,000 readings in Python and found the canteen PM2.5 spike at lunch.

Data Science

Computing

Student-added

Computing



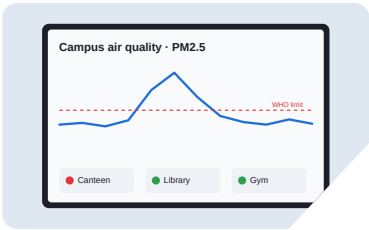
Youth hackathon: 2nd place

28 Sep 2026

Team of three built a flood-alert chatbot in 24 hours.

Computing

School-verified `NIST-1T6TUH`



Air-quality data report

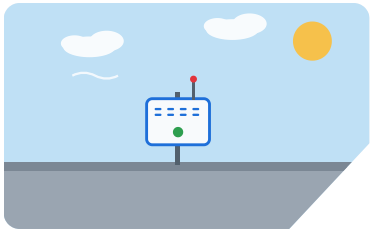
28 Mar 2026

Cleaned 40,000 readings in Python and found the canteen PM2.5 spike at lunch.

Data Science

Computing

Student-added



Air-quality sensor network

14 Mar 2026

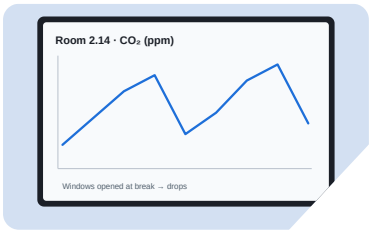
Built 6 Raspberry Pi sensors around campus and a live dashboard.

> *"Maya designed the sensor housing, calibrated every unit against the school weather station and wrote the dashboard herself. Outstanding independent work."*
>
> **Ms Priya Nair, Computer Science**

Computing

Environment

Teacher-endorsed `NIST-1TGBH2`



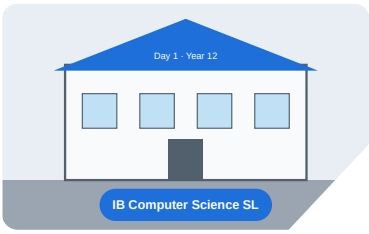
CO_2 sensor prototype

5 Nov 2025

One classroom sensor on a Raspberry Pi, logging every minute.

Computing

Student-added



Started IB Computer Science SL

21 Aug 2025

Depot: IB CS SL with Ms Nair.

Computing

School-verified `NIST-1U0VT2`



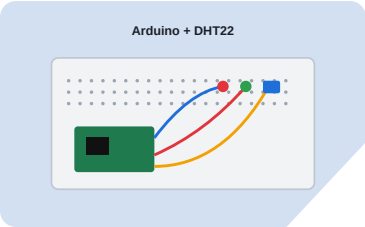
Data visualisation course

10 May 2025

Online course in pandas and matplotlib; 8 graded notebooks.

Computing

Student-added



Arduino workshop

5 Oct 2024

Wired a temperature sensor and logged readings to a laptop.

Computing

School-verified `NIST-PI8DA8`



Recycling quiz web app

5 Mar 2024

First web app in HTML and JavaScript, used in a Y7 assembly.

Computing

Student-added

OTHER CONTEXT

Environment · 6 Service · 5

School-verified Teacher-endorsed Student-added

Every item shows how it was verified. Students choose and order items but cannot edit verified facts.

NIST International School

Maya Chen · NIST Transit Portfolio

Printed 27 October 2026 from a share link. Scan the code or go to **localhost:4961/#/s/tp-nb7d** to read the live portfolio while the link is open.

Each verified item carries a code starting NIST-. Anyone can check a code at **localhost:4961/#/v**. The school confirms what came from its systems or was endorsed by a teacher.